STK4021 – Mandatory assignment, Fall 2026

Bayesian inversion of seismic data

NAME – STK4021

14 October 2026

Self-report

To be written by the student: how the work proceeded, what was difficult, time-consuming or fun, what was relatively easy, collaboration and sources used (including AI tools).

Data and forward operator

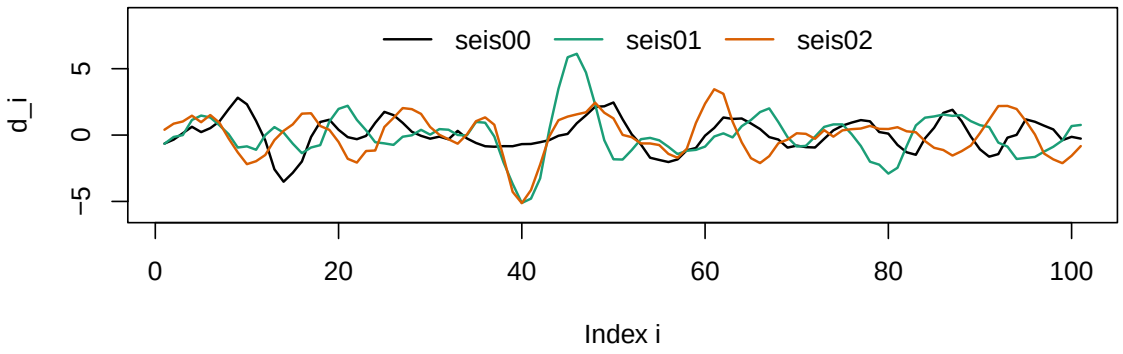


Figure 1: The three observed seismic traces.

The forward operator $\mathbf{A}$ is a 101×101 matrix. Every row holds the same wavelet, shifted one sample, so d_i depends on $\mathbf{m}$ only through values near index i. One property is important for the interpretation. The rows of $\mathbf{A}$ sum to zero, so $\mathbf{A1} = \mathbf{0}$ (numerically, $\max_i |(\mathbf{A1})_i| = 6 \times 10^{-16}$). Hence $\mathbf{A}$ has a zero singular value (the smallest is 1.1×10^{-15}, the largest 2.85). A constant shift of the whole profile does not change the data, so it cannot be identified from the data. The data only carry information about *changes* in $\mathbf{m}$.

Part 1: Gaussian Bayesian inversion and simulation

1a) Joint distribution and posterior

The model is $\mathbf{d} = \mathbf{Am} + \boldsymbol{\varepsilon}$ with $\mathbf{m} \sim N(\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)$, $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I})$, and $\mathbf{m}$ and $\boldsymbol{\varepsilon}$ independent. Because of independence, $(\mathbf{m}, \boldsymbol{\varepsilon})$ is jointly Gaussian with mean $(\boldsymbol{\mu}_m, \mathbf{0})$ and block-diagonal covariance $\mathrm{diag}(\boldsymbol{\Sigma}_m, \sigma_\varepsilon^2 \mathbf{I})$. The pair $(\mathbf{m}, \mathbf{d})$ is a linear transformation of this vector,

$$\begin{pmatrix} \mathbf{m} \\ \mathbf{d} \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{m} \\ \boldsymbol{\varepsilon} \end{pmatrix},$$

so $(\mathbf{m}, \mathbf{d})$ is also jointly Gaussian. Its moments follow from linearity of expectation and the rules for covariances of linear combinations:

$$\begin{aligned} E(\mathbf{d}) &= \mathbf{A}E(\mathbf{m}) + E(\boldsymbol{\varepsilon}) = \mathbf{A}\boldsymbol{\mu}_m, \\ \mathrm{Cov}(\mathbf{d}) &= \mathbf{A}\,\mathrm{Cov}(\mathbf{m})\mathbf{A}^\top + \mathrm{Cov}(\boldsymbol{\varepsilon}) = \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I}, \\ \mathrm{Cov}(\mathbf{d}, \mathbf{m}) &= \mathrm{Cov}(\mathbf{Am} + \boldsymbol{\varepsilon}, \mathbf{m}) = \mathbf{A}\,\mathrm{Cov}(\mathbf{m}, \mathbf{m}) + \mathrm{Cov}(\boldsymbol{\varepsilon}, \mathbf{m}) = \mathbf{A}\boldsymbol{\Sigma}_m, \end{aligned}$$

where $\mathrm{Cov}(\boldsymbol{\varepsilon}, \mathbf{m}) = \mathbf{0}$ by independence, and $\mathrm{Cov}(\mathbf{m}, \mathbf{d}) = \mathrm{Cov}(\mathbf{d}, \mathbf{m})^\top = \boldsymbol{\Sigma}_m\mathbf{A}^\top$. Hence

$$\begin{pmatrix} \mathbf{m} \\ \mathbf{d} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_m \\ \mathbf{A}\boldsymbol{\mu}_m \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_m\mathbf{A}^\top \\ \mathbf{A}\boldsymbol{\Sigma}_m & \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I} \end{pmatrix} \right).$$

In particular, the marginal distribution of the data is $\mathbf{d} \sim N(\mathbf{A}\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_d)$ with $\boldsymbol{\Sigma}_d = \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I}$. We use this again in Part 2.

Posterior. For a partitioned Gaussian vector, $\boldsymbol{Y}_1 \mid \boldsymbol{Y}_2 = \boldsymbol{y}_2 \sim N\big(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{y}_2 - \boldsymbol{\mu}_2),\ \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}\big)$. With $\boldsymbol{Y}_1 = \mathbf{m}$ and $\boldsymbol{Y}_2 = \mathbf{d}$ this gives

$$p(\mathbf{m} \mid \mathbf{d}) = N(\mathbf{m}; \boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d}),$$

$$\begin{aligned}
\boldsymbol{\mu}_{m|d} &= \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_m \mathbf{A}^\top \big(\mathbf{A}\boldsymbol{\Sigma}_m \mathbf{A}^\top + \sigma_\varepsilon^2 \mathbf{I}\big)^{-1} (\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m), \\
\boldsymbol{\Sigma}_{m|d} &= \boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}_m \mathbf{A}^\top \big(\mathbf{A}\boldsymbol{\Sigma}_m \mathbf{A}^\top + \sigma_\varepsilon^2 \mathbf{I}\big)^{-1} \mathbf{A}\boldsymbol{\Sigma}_m.
\end{aligned}$$

The posterior mean is the prior mean plus a correction proportional to the data residual $\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m$. The posterior covariance is the prior covariance minus what the data explain, and it does not depend on the observed values of $\mathbf{d}$.

Each marginal of a Gaussian vector is Gaussian, so $m_i \mid \mathbf{d} \sim N\big((\boldsymbol{\mu}_{m|d})_i, (\boldsymbol{\Sigma}_{m|d})_{ii}\big)$. The pointwise 95% credibility interval used below is therefore $(\boldsymbol{\mu}_{m|d})_i \pm 1.96\sqrt{(\boldsymbol{\Sigma}_{m|d})_{ii}}$, and the same holds for the prior.

1b) Prior: mean, 95% interval and simulations

We use $\mu_{m,i} = \mu$ and $\Sigma_{m,ij} = \sigma_m^2 \exp(-3|i-j|/\ell)$ with $(\mu, \sigma_m^2, \ell, \sigma_\varepsilon^2) = (0, 1, 20, 0.04)$. The prior variance is $\Sigma_{m,ii} = \sigma_m^2 = 1$ for every i, so the prior expectation is 0 and the 95% credibility interval is 0 ± 1.96 for every component. The correlation between m_i and m_j is $\exp(-3|i-j|/20)$. It is 0.86 for neighbours, 0.22 at distance 10, and $e^{-3} \approx 0.05$ at distance $\ell = 20$.

Realisations are simulated with the linear-transform rule: if $\boldsymbol{x} \sim N(\mathbf{0}, \mathbf{I})$ and $\mathbf{L}\mathbf{L}^\top = \boldsymbol{\Sigma}_m$ (Cholesky factorisation), then $\boldsymbol{\mu}_m + \mathbf{L}\boldsymbol{x} \sim N(\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)$.

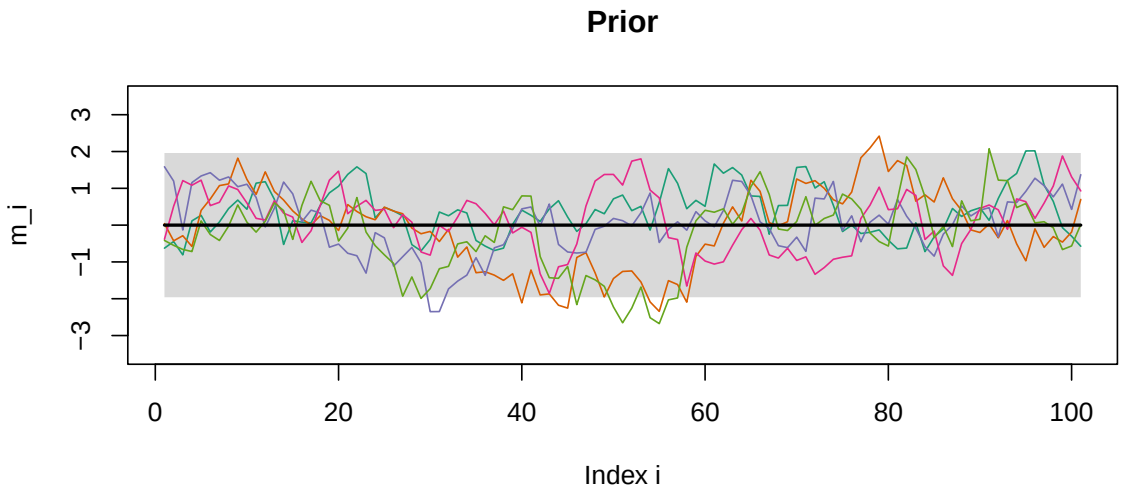


Figure 2: Prior expectation (black), pointwise 95% credibility interval (grey) and five simulated prior profiles.

The simulated profiles are irregular from sample to sample, which is typical of the exponential correlation function. They stay correlated over roughly 10–20 samples and wander around 0 within about ± 2, as the prior specifies.

1c) Posterior given `seis00`

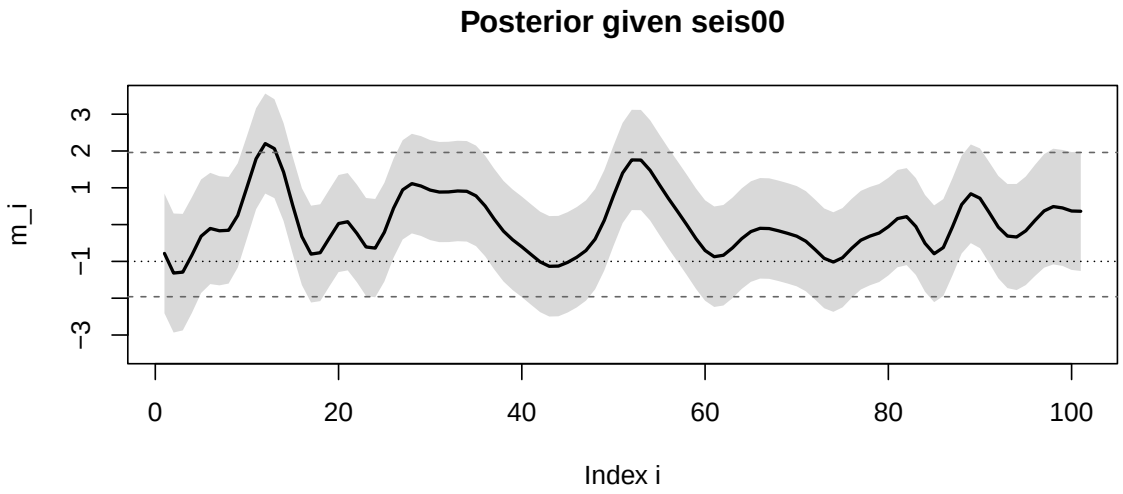


Figure 3: Posterior expectation (black) and pointwise 95% credibility interval (grey) given seis00. Dashed lines: prior 95% limits; dotted line: the level $c = -1$ used in 1d.

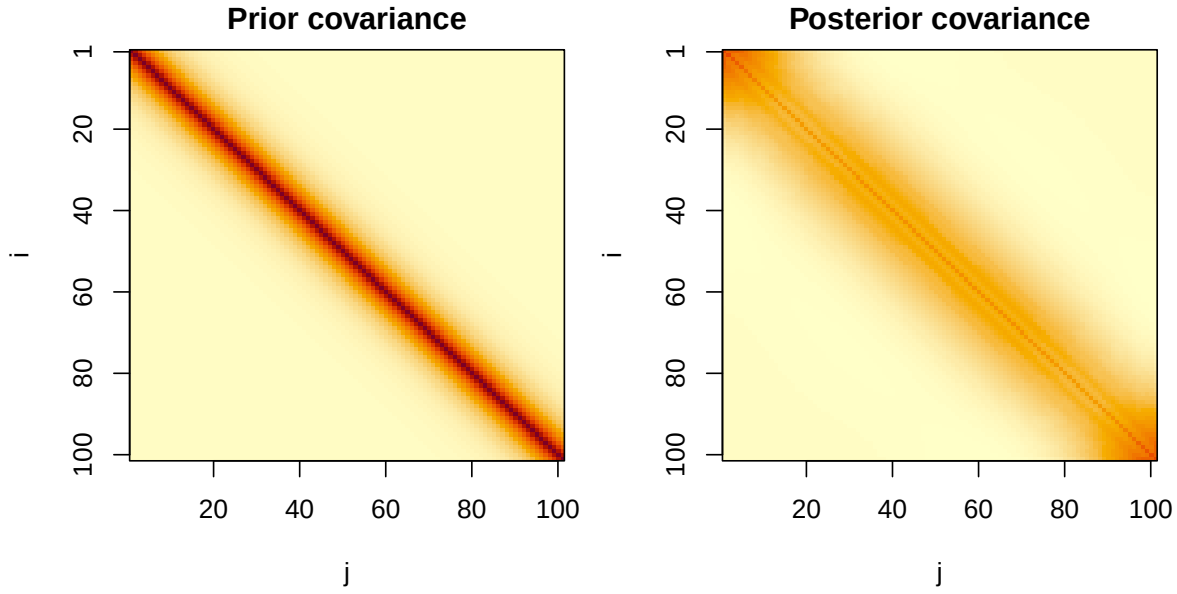


Figure 4: Prior (left) and posterior (right) covariance matrices on the same colour scale. Row index i downwards, column index j to the right.

Check of the computation. The posterior can also be written in the parameter-space (precision) form $\boldsymbol{\Sigma}_{m|d}^{-1} = \boldsymbol{\Sigma}_m^{-1} + \mathbf{A}^\top\mathbf{A}/\sigma_\varepsilon^2$, $\boldsymbol{\mu}_{m|d} = \boldsymbol{\Sigma}_{m|d}\big(\boldsymbol{\Sigma}_m^{-1}\boldsymbol{\mu}_m + \mathbf{A}^\top\mathbf{d}/\sigma_\varepsilon^2\big)$. The two forms agree to within 2.3×10^{-13}, which confirms the computation.

Interpretation.

- **Mean.** The posterior expectation follows the data. It is pulled below zero where the trace indicates a decrease in $\mathbf{m}$, and it falls below $c = -1$ only at indices 2, 3, 42, 43, 44, 45, 74.
- **Variance.** The posterior standard deviations lie between 0.67 and 0.83, compared with 1 in the prior. In the interior the variance is roughly halved. It is larger near the two ends, where fewer rows of $\mathbf{A}$ involve the parameters.
- **Correlation.** The covariance images show a second effect. In the prior, the covariance is concentrated close to the diagonal. In the posterior, it is lower on the diagonal but spread further out. For $i = 51$, the posterior correlation with m_{i+1}, m_{i+10} and m_{i+20} is 0.8, 0.67, 0.32, compared with 0.86, 0.22, 0.05 in the prior.
- **Why.** This follows from $\mathbf{A1} = \mathbf{0}$ and the small singular values of $\mathbf{A}$. The data determine local changes in $\mathbf{m}$ well, but a direction in parameter space with zero singular value does not affect the data and keeps its prior uncertainty. The variance of the profile average $\bar{m} = \frac{1}{n}\sum_i m_i$ is 0.124 in the prior and 0.121 in the posterior, so it is essentially not reduced. What remains is uncertainty in the overall level, shared by all components. That shared uncertainty is the slowly decaying positive correlation in the posterior.

1d) Exceedance volume

$V = h(\mathbf{m}) = \sum_{i=1}^{n} |m_i - c|\, I(m_i < c) = \sum_{i=1}^{n} (c - m_i)\, I(m_i < c)$ with $c = -1$ is a non-linear function of $\mathbf{m}$, and its posterior has no closed form. We therefore use Monte Carlo. We draw $\mathbf{m}^{(1)}, \ldots, \mathbf{m}^{(B)}$ independently from $N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$ (as in 1b, with the posterior Cholesky factor) and compute $V^{(b)} = h(\mathbf{m}^{(b)})$. Then $V^{(1)}, \ldots, V^{(B)}$ is an independent sample from $p(V \mid \mathbf{d})$.

Table 1: Posterior summaries of V from $B = 100$ samples.

mean	median	sd
8.76	5.48	8.68

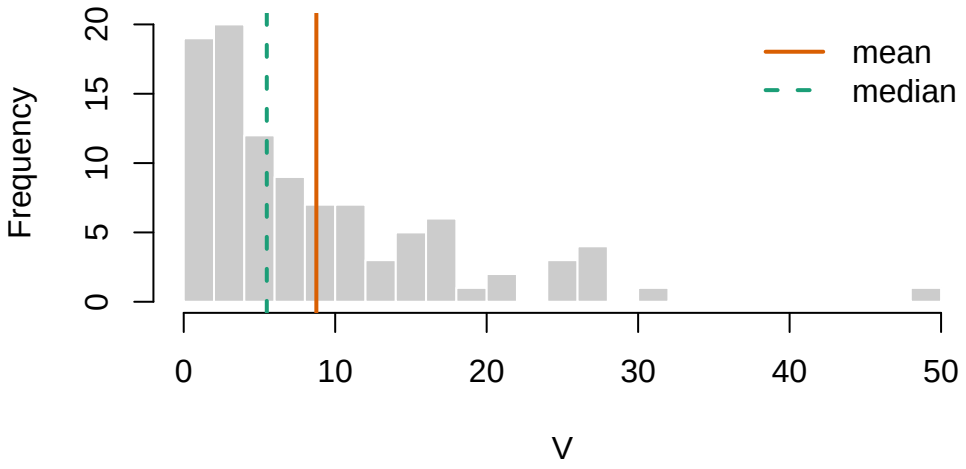


Figure 5: Histogram of the $B = 100$ posterior samples of V, with the sample mean and median.

The posterior of V is strongly right-skewed. Most posterior profiles go only slightly below c, while a few have

long or deep excursions and give large V. The mean (8.76) is therefore larger than the median (5.48), and the standard deviation (8.68) is as large as the mean, so V is very uncertain. 2 of the 100 samples have $V = 0$. Although the posterior expectation is above c at almost all indices, nearly every posterior profile goes below c somewhere. The posterior mean profile is much smoother than any single posterior realisation, and h applied to it gives only $h(\boldsymbol{\mu}_{m|d}) = 0.93$, far below the posterior mean of V. Since h is non-linear, $E\{h(\mathbf{m}) \mid \mathbf{d}\} \neq h(E\{\mathbf{m} \mid \mathbf{d}\})$, which is why V has to be computed from simulated profiles.

1e) Percentiles and their uncertainty

Estimates. Among $B = 100$ sorted samples $V_{(1)} \leq \cdots \leq V_{(100)}$, a fraction $k/100$ lies at or below $V_{(k)}$. The natural estimates are therefore $\widehat{P}_{10} = V_{(10)} = 0.92$ and $\widehat{P}_{90} = V_{(90)} = 20.64$. So about 80% of the posterior probability for V lies between 0.9 and 20.6.

Uncertainty in P_{10}. Let N be the number of the B samples that are below the true P_{10}. The samples are independent and each falls below P_{10} with probability 0.1, so $N \sim \mathrm{Bin}(B, 0.1)$. The true P_{10} lies between the r-th and s-th order statistics exactly when between r and $s-1$ samples fall below it:

$$P\big(V_{(r)} \leq P_{10} < V_{(s)}\big) = P(r \leq N \leq s-1) = \sum_{k=r}^{s-1} \binom{B}{k} 0.1^k \, 0.9^{B-k}.$$

This only requires the samples to be independent and $p(V \mid \mathbf{d})$ to be continuous around P_{10}. Here V has a small probability mass at $V = 0$, but $P_{10} > 0$, so the condition holds, and no assumption about the shape of $p(V \mid \mathbf{d})$ is needed. We take r and $s-1$ as the 2.5% and 97.5% quantiles of $\mathrm{Bin}(B, 0.1)$, which gives about 95% probability.

Table 2: Estimate of P_{10} and the order-statistic interval $[V_{(r)}, V_{(s)}]$, with the binomial probability that it contains P_{10}.

B	estimate of P10	r	s	V_(r)	V_(s)	probability
100	0.92	5	17	0.19	1.82	0.956
10000	0.99	942	1060	0.93	1.06	0.951

With $B = 100$ the interval is $[V_{(5)}, V_{(17)}] = [0.19,\ 1.82]$. That is wide compared with the estimate itself, so 100 samples are not enough to pin down P_{10} (or P_{90}). With $B = 10\,000$, the same construction gives the much narrower interval $[0.93,\ 1.06]$. The width of the interval shrinks roughly like $1/\sqrt{B}$, because the standard deviation of N/B is $\sqrt{0.1 \cdot 0.9/B}$.

1f) Permutation test of exchangeability

If $d_1, \ldots, d_n$ are exchangeable, their joint distribution is unchanged by any permutation of the indices. Under this hypothesis, the observed ordering is just one of the $n!$ equally likely orderings. The value of any statistic for the observed ordering is then a typical draw from the distribution of the statistic over random permutations. As the statistic we use the lag-1 correlation

$$\hat{\rho}_1 = \mathrm{cor}\big((d_1, \ldots, d_{n-1}), (d_2, \ldots, d_n)\big),$$

which measures dependence between neighbouring samples. Under exchangeability it should be close to the values from permuted data.

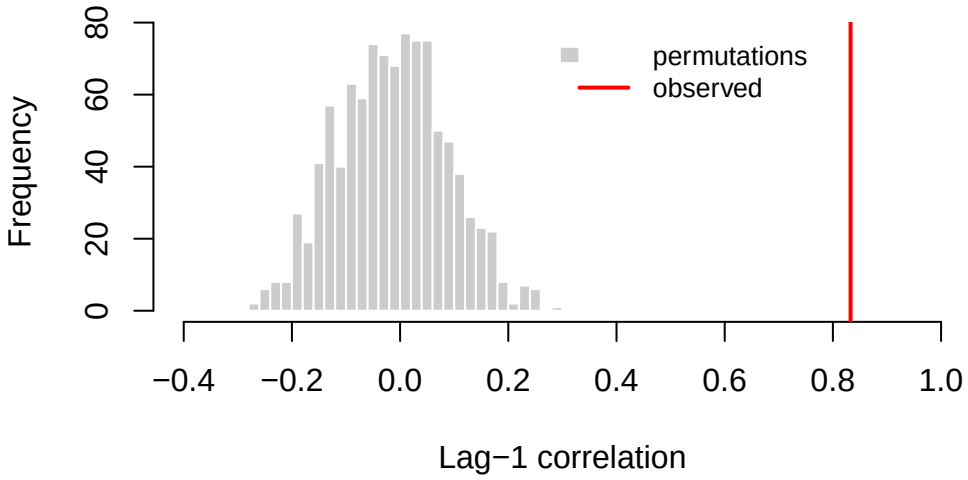


Figure 6: Lag-1 correlation for 1000 random permutations of seis00 (histogram) and for the observed ordering (red line).

The observed lag-1 correlation of seis00 is $\hat{\rho}_1 = 0.833$. The 1000 permuted values are centred close to zero (mean -0.013) and range from -0.28 to 0.29. None of the 1000 permutations gives a correlation as extreme as the observed one, so the estimated p-value is below 1/1000. The claim that the data are exchangeable is clearly rejected. (The other two traces give the same picture, with lag-1 correlations 0.84 and 0.83.)

This agrees with our model. For exchangeable variables every pair (d_i, d_j), $i \neq j$, has the same joint distribution and hence the same covariance. Under $\mathbf{d} = \mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}$, the covariance matrix $\mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I}$ is not of that form: neighbouring samples share the wavelet and parts of $\mathbf{m}$ and are strongly correlated, while distant samples are nearly uncorrelated. A model that treats the seismic samples as exchangeable (for example iid given some parameter) is therefore wrong for these data. The ordering of the samples carries the information about the subsurface.

Appendix: R code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE,
                      fig.align = "center", fig.width = 7.5, fig.height = 3.6,
                      out.width = "95%", fig.pos = "H")
A  <- as.matrix(read.table("data/A.txt"))   # forward operator (n x n)
dimnames(A) <- NULL
d0 <- scan("data/seis00.txt", quiet = TRUE) # seismic traces
d1 <- scan("data/seis01.txt", quiet = TRUE)
d2 <- scan("data/seis02.txt", quiet = TRUE)
n  <- nrow(A)
z  <- 1:n                                   # index along the profile

A_rowsum_max <- max(abs(A %*% rep(1, n)))   # check: A applied to a constant vector
A_sv         <- svd(A)$d                    # singular values of A
matplot(z, cbind(d0, d1, d2), type = "l", lty = 1, lwd = 1.5,
        col = c("black", "#1b9e77", "#d95f02"), xlab = "Index i", ylab = "d_i",
        ylim = c(-6, 9))
legend("top", c("seis00", "seis01", "seis02"), lty = 1, lwd = 1.5,
       col = c("black", "#1b9e77", "#d95f02"), bty = "n", horiz = TRUE)
# Prior covariance: exponential correlation function
exp_cov <- function(n, sigma2, ell) {
  D <- abs(outer(1:n, 1:n, "-"))           # matrix of |i - j|
  sigma2 * exp(-3 * D / ell)
}

# Posterior of m | d for d = A m + eps (Gaussian conditioning, data-space form)
gauss_posterior <- function(d, A, mu, Sigma, sigma2_eps) {
  Sigma_d <- A %*% Sigma %*% t(A) + sigma2_eps * diag(nrow(A))   # Cov(d)
  K       <- Sigma %*% t(A) %*% solve(Sigma_d)                   # Cov(m,d) Cov(d)^-1
  mu_post <- drop(mu + K %*% (d - A %*% mu))
  S_post  <- Sigma - K %*% A %*% Sigma
  S_post  <- (S_post + t(S_post)) / 2      # make exactly symmetric (rounding)
  list(mean = mu_post, cov = S_post)
}

# Simulate B draws from N(mu, S): m = mu + L x, L L^T = S, x ~ N(0, I)
rmvn <- function(B, mu, S) {
  L <- t(chol(S + 1e-10 * diag(length(mu))))   # lower Cholesky factor (tiny jitter)
  X <- matrix(rnorm(length(mu) * B), length(mu), B)
  t(mu + L %*% X)                              # one draw per row
}

# Mean with pointwise 95% credibility band (and optional realisations)
band_plot <- function(mean, sd, main, ylim, sims = NULL) {
  lo <- mean - 1.96 * sd
  hi <- mean + 1.96 * sd
  plot(z, mean, type = "n", ylim = ylim, xlab = "Index i", ylab = "m_i", main = main)
  polygon(c(z, rev(z)), c(lo, rev(hi)), col = "grey85", border = NA)
  if (!is.null(sims)) matlines(z, t(sims), lty = 1, lwd = 1,
                               col = c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e"))
  lines(z, mean, lwd = 2)
}
```

```r
mu <- 0; sigma2_m <- 1; ell <- 20; sigma2_eps <- 0.04
mu_m    <- rep(mu, n)
Sigma_m <- exp_cov(n, sigma2_m, ell)

set.seed(1)
prior_sims <- rmvn(5, mu_m, Sigma_m)
band_plot(mu_m, sqrt(diag(Sigma_m)), "Prior", ylim = c(-3.5, 3.5), sims = prior_sims)
post0 <- gauss_posterior(d0, A, mu_m, Sigma_m, sigma2_eps)
sd0   <- sqrt(diag(post0$cov))
band_plot(post0$mean, sd0, "Posterior given seis00", ylim = c(-3.5, 3.5))
abline(h = c(-1.96, 1.96), lty = 2, col = "grey40")
abline(h = -1, lty = 3)
cov_image <- function(S, main, zlim) {
  image(z, z, t(S)[, n:1], zlim = zlim, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "j", ylab = "i", main = main, axes = FALSE, useRaster = TRUE)
  axis(1); axis(2, at = n + 1 - c(1, 20, 40, 60, 80, 100), labels = c(1, 20, 40, 60, 80, 100))
  box()
}
zl <- range(Sigma_m, post0$cov)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
cov_image(Sigma_m, "Prior covariance", zl)
cov_image(post0$cov, "Posterior covariance", zl)
# Check 1: the parameter-space (precision) form gives the same posterior
Q_post    <- solve(Sigma_m) + t(A) %*% A / sigma2_eps
S_check   <- solve(Q_post)
mu_check  <- S_check %*% (solve(Sigma_m) %*% mu_m + t(A) %*% d0 / sigma2_eps)
check_err <- max(abs(S_check - post0$cov), abs(mu_check - post0$mean))

# Numbers used in the discussion
sd_range  <- range(sd0)
mid       <- 51
cor_post  <- post0$cov[mid, mid + c(1, 10, 20)] / post0$cov[mid, mid]
cor_prior <- exp(-3 * c(1, 10, 20) / ell)
w         <- rep(1 / n, n)                  # weights giving the profile average
var_avg   <- c(prior = drop(t(w) %*% Sigma_m %*% w),
               post  = drop(t(w) %*% post0$cov %*% w))
low_idx   <- which(post0$mean < -1)
c_lev <- -1
h_V   <- function(m, c = c_lev) sum((c - m) * (m < c))   # exceedance volume

B <- 100
set.seed(2)
post_sims <- rmvn(B, post0$mean, post0$cov)
V <- apply(post_sims, 1, h_V)                              # one V per posterior draw

V_summary <- c(mean = mean(V), median = median(V), sd = sd(V))
knitr::kable(t(round(V_summary, 2)),
             caption = "Posterior summaries of $V$ from $B=100$ samples.")
hist(V, breaks = 20, col = "grey80", border = "white", main = "", xlab = "V")
abline(v = V_summary["mean"],   lty = 1, lwd = 2, col = "#d95f02")
abline(v = V_summary["median"], lty = 2, lwd = 2, col = "#1b9e77")
legend("topright", c("mean", "median"), lty = 1:2, lwd = 2,
       col = c("#d95f02", "#1b9e77"), bty = "n")
```

```r
Vs  <- sort(V)                                        # order statistics V_(1) <= ... <= V_(B)
P10 <- Vs[10]
P90 <- Vs[90]

# N = number of samples below the true P10 ~ Bin(B, 0.1).
# V_(r) <= P10 < V_(s)  <=>  r <= N <= s - 1.
r <- qbinom(0.025, B, 0.1)
s <- qbinom(0.975, B, 0.1) + 1
coverage <- pbinom(s - 1, B, 0.1) - pbinom(r - 1, B, 0.1)

# Same construction for a larger sample, to judge whether B = 100 is enough
B_big <- 10000
set.seed(3)
V_big  <- apply(rmvn(B_big, post0$mean, post0$cov), 1, h_V)
Vs_big <- sort(V_big)
r_big  <- qbinom(0.025, B_big, 0.1)
s_big  <- qbinom(0.975, B_big, 0.1) + 1
cov_big <- pbinom(s_big - 1, B_big, 0.1) - pbinom(r_big - 1, B_big, 0.1)

tab_P10 <- data.frame(
  B        = c(B, B_big),
  estimate = c(P10, Vs_big[B_big / 10]),
  r        = c(r, r_big),
  s        = c(s, s_big),
  lower    = c(Vs[r], Vs_big[r_big]),
  upper    = c(Vs[s], Vs_big[s_big]),
  coverage = c(coverage, cov_big))
knitr::kable(tab_P10, digits = c(0, 2, 0, 0, 2, 2, 3),
             col.names = c("B", "estimate of P10", "r", "s",
                           "V_(r)", "V_(s)", "probability"),
             caption = "Estimate of $P_{10}$ and the order-statistic interval $[V_{(r)},V_{(s)}]$, with
lag1 <- function(x) cor(x[-1], x[-length(x)])   # lag-1 correlation

rho_obs <- lag1(d0)
set.seed(4)
rho_perm <- replicate(1000, lag1(sample(d0)))    # sample() gives a random permutation
n_extreme <- sum(abs(rho_perm) >= abs(rho_obs))

hist(rho_perm, breaks = 30, col = "grey80", border = "white", main = "",
     xlab = "Lag-1 correlation", xlim = c(-0.4, 1))
abline(v = rho_obs, col = "red", lwd = 2)
legend(x = 0.25, y = par("usr")[4], c("permutations", "observed"), cex = 0.85, fill = c("grey80", NA),
       border = c("white", NA), lty = c(NA, 1), lwd = c(NA, 2),
       col = c(NA, "red"), bty = "n")
```